

Proportions

Two equal ratios

1 A proportion

A diagram showing the proportion $\frac{a}{b} = \frac{c}{d}$. The variable a is blue, b is green, c is red, and d is purple. Two arrows cross each other: a blue arrow from a to d and a red arrow from c to b , illustrating the cross-multiplication process.

$$a \times d = b \times c$$

Cross products are equal in a proportion.

2 Solve for a missing value

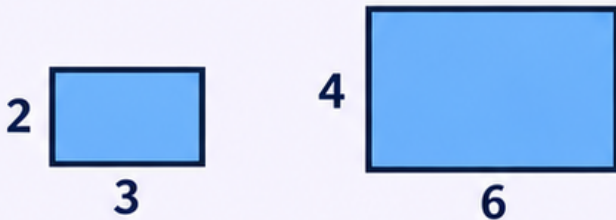
$$\frac{3}{4} = \frac{x}{12}$$

$$4x = 36$$

$$x = 9$$

Cross-multiply, then solve for the variable.

3 Same ratio, different size



The side lengths are proportional:

$$\frac{2}{3} = \frac{4}{6}$$

Same shape, different size, same ratio.

4 Key ideas



A proportion sets two ratios equal.



Use cross products:
 $a \times d = b \times c$.



Solve for the missing value.



Proportions show equivalent quantities, even if the size changes.



Cross-multiply, then solve for the missing value.

Proportions

Two equal ratios

1 A proportion

$$\frac{a}{b} = \frac{c}{d}$$

$$a \times d = b \times c$$

Cross products are equal
in a proportion.

2 Solve for a missing value

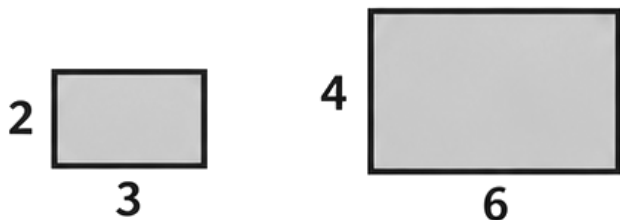
$$\frac{3}{4} = \frac{x}{12}$$

$$4x = 36$$

$$x = 9$$

Cross-multiply, then solve
for the variable.

3 Same ratio, different size



The side lengths are proportional:

$$\frac{2}{3} = \frac{4}{6}$$

Same shape, different size,
same ratio.

4 Key ideas

- ✓ A proportion sets two ratios equal.
- ✓ Use cross products:
 $a \times d = b \times c$.
- ✓ Solve for the missing value.
- ✓ Proportions show equivalent quantities, even if the size changes.



Cross-multiply, then solve for the missing value.